

YouTube Viral Video Prediction

Using Pre-Publication Information

Yale University
CPSC 3180/5180: Introduction to Machine Learning
Spring 2026

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Abstract

This project investigates whether a YouTube video’s virality can be predicted using only information available before publication: the video title, tags, publish timestamp, content category, and description length. We define a video as viral if its view count reaches the 80th percentile among all videos in the dataset, yielding a binary classification task with roughly 20% positive examples. Two models are compared: a Logistic Regression baseline trained on nine structured features, and an RBF Kernel SVM trained on the same structured features augmented with 384-dimensional sentence embeddings of the video title and tags. The SVM achieves an AUC of 0.711 and an F1 score of 0.434, compared to 0.599 and 0.352 for the baseline, suggesting that semantic content provides meaningful additional predictive signal. Analysis of the Logistic Regression coefficients reveals that tag count and the presence of exclamation marks in the title are strong positive predictors, while question marks are negative, suggesting that emotionally-driven titles are associated with higher viral rates than informational ones. A cross-category generalization experiment further reveals that viral patterns are strongly category-specific, with in-domain F1 averaging 0.859 and cross-domain F1 averaging 0.166, a finding that points toward category-aware modeling as a natural direction for future work.

Keywords: viral prediction, sentence embeddings, logistic regression, kernel SVM, PCA

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1 Introduction

Predicting whether a YouTube video will go viral is a problem of practical interest to content creators, platform operators, and marketers. Most approaches to this problem rely on post-publication engagement signals such as view counts, likes, and comment volume — metrics that are only observable after a video has already been released and seen by an audience. This project poses a stricter and more realistic question: can we predict virality using only information that exists at the moment of upload?

The features considered are limited to the video title, tags, publish timestamp, content category, and description length, all of which are determined before a video goes live. Engagement metrics (views, likes, dislikes, comment count) are explicitly excluded as model inputs; view counts are used solely to construct the binary viral label.

Beyond the core prediction task, the project investigates a second question: do viral patterns generalize across content categories? A model trained on Entertainment videos may have learned patterns specific to that genre. Testing whether those patterns transfer to Music or Comedy reveals whether virality is a universal phenomenon or a category-specific one — a question with direct implications for how such models should be designed and deployed in practice.

Two main experiments are conducted. Experiment 1 trains a Logistic Regression classifier on structured features only to establish a baseline and to identify which individual variables carry the most predictive weight. Experiment 2 augments these features with sentence embeddings of the video title and tags, encoding their semantic content into a dense 384-dimensional vector, and trains an RBF Kernel SVM to test whether the meaning of a title carries additional predictive power beyond the structured features. Two further experiments visualize the learned feature space via PCA and measure cross-category generalization through a held-out F1 matrix.

2 Data

2.1 Dataset

The dataset used is the **Kaggle Trending YouTube Video Statistics** (US subset), consisting of a CSV file of video records and a JSON category mapping file. It contains metadata for videos that appeared on YouTube’s US trending page, including the video title, tags, publish timestamp, category identifier, description text, view count, like count, dislike count, and comment count. The raw dataset contains 40,949 rows. Because the same video can appear on multiple trending days, we deduplicate by video identifier and retain only the first appearance of each video, reducing the dataset to 6,351 unique videos. After removing rows with missing titles or unparseable publish timestamps, all 6,351 samples are retained.

2.2 Label Definition

A binary viral label is constructed from view counts:

$$\text{viral} = \begin{cases} 1 & \text{if views} \geq Q_{0.80}(\text{views}) \\ 0 & \text{otherwise} \end{cases}$$

where $Q_{0.80}$ denotes the 80th percentile of the view count distribution (933,053 views). This yields approximately 20% positive (viral) examples, providing a more balanced class distribution for model training.

2.3 Category Distribution

The viral rate varies substantially across content categories, as shown in Figure 1. Gaming leads with a 40.8% viral rate, while News & Politics and Pets & Animals fall below 6%. This strong category-level variation suggests that content category is an important predictor, a hypothesis confirmed by the feature importance analysis in Section 5.

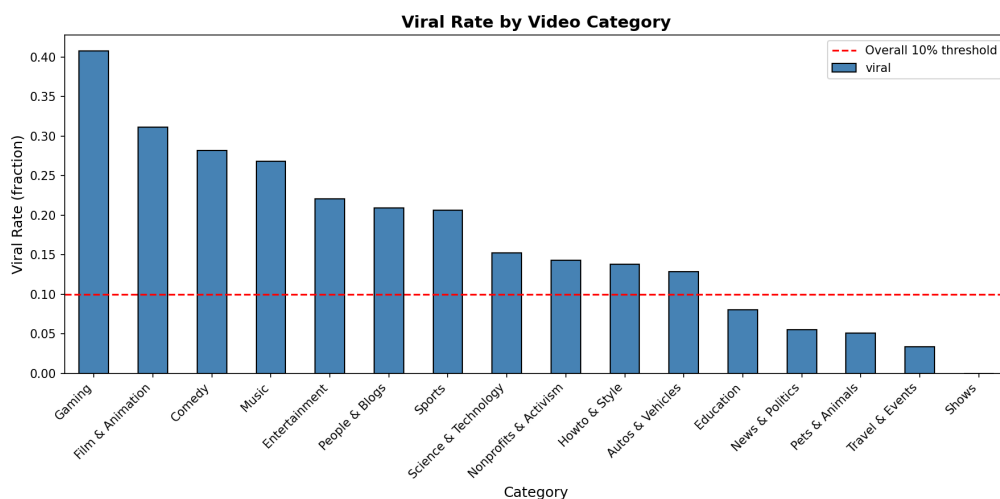


Figure 1: Viral rate broken down by YouTube content category. Gaming exhibits the highest viral rate at 40.8%, while News & Politics and Pets & Animals fall below 6%.

3 Methodology

3.1 Feature Engineering

Two feature sets are constructed from the allowed pre-publication variables.

Structured features (9 dimensions):

- publish hour — hour of day (0–23) extracted from the publish timestamp
- publish weekday — day of week (0 = Monday, 6 = Sunday)
- title length — number of characters in the title

- title word count — number of whitespace-delimited tokens in the title
- title has exclamation — binary indicator for the presence of “!”
- title has question — binary indicator for the presence of “?”
- tag count — number of tags associated with the video
- description length — character length of the video description
- category id — integer category identifier

Semantic embedding features: To capture the meaning of the video title and tags, we concatenate the title text with the cleaned tag string and encode the result using the pretrained sentence transformer model *all-MiniLM-L6-v2*, producing a 384-dimensional dense embedding per video. Unlike bag-of-words approaches such as TF-IDF, sentence embeddings capture semantic similarity and contextual meaning, allowing the model to recognize that “Epic Fails Compilation” and “Best Bloopers Ever” carry similar signals even without lexical overlap.

The full feature matrix concatenates the scaled structured features and the sentence embedding, producing a 393-dimensional dense representation per video. To prevent data leakage, the standard scaler for structured features and the PCA transform used in Experiment 2 are fitted exclusively on the training split and then applied to the test split. The sentence transformer model is pretrained and frozen; it is not fine-tuned on this dataset.

3.2 Train/Test Split

The dataset is divided into a 70% training set (4,445 samples) and a 30% test set (1,906 samples) using stratified sampling (preserving the viral rate in both splits) with a fixed random seed for reproducibility.

3.3 Experiment 1 — Logistic Regression Baseline

Logistic Regression with ℓ_2 regularization is trained on the nine structured features only. The model minimizes the regularized cross-entropy loss:

$$\min_{w,b} \sum_{i=1}^n \log(1 + e^{-y_i(w^\top x_i + b)}) + \frac{1}{2C} \|w\|^2$$

where $C > 0$ controls regularization strength. Class imbalance is handled by reweighting each training example inversely proportional to its class frequency. The optimal C is selected via 5-fold stratified cross-validation over the grid $C \in \{0.001, 0.01, 0.1, 1, 10\}$, with F1 score as the selection criterion. Because Logistic Regression produces interpretable coefficients, this experiment also serves as a feature importance analysis.

3.4 Experiment 2 — RBF Kernel SVM

A Support Vector Machine with the Radial Basis Function (RBF) kernel is trained on the full 393-dimensional feature set. To improve computational efficiency, PCA first reduces the feature matrix to 100 principal components (retaining 97.33% of total variance). The RBF kernel implicitly maps inputs into a high-dimensional feature space:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

The SVM then finds the maximum-margin separating hyperplane in this space. Hyperparameters $C \in \{0.1, 1, 10\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$ are jointly tuned via 5-fold cross-validation optimizing F1. Class imbalance is handled identically to Experiment 1.

3.5 Experiment 3 — PCA Visualization

To examine the geometry of the feature space, PCA is applied to project the full training feature matrix to 2 dimensions. Viral and non-viral samples are plotted separately to assess visual separability, both for the full training set and individually for four representative categories: Gaming, Music, Entertainment, and Science & Technology. These four categories are selected to maximize visual contrast — Gaming has the highest viral rate (40.8%) and Music the lowest cross-domain transfer, making them informative anchors for understanding the feature space. Note that Experiment 4 instead uses the four highest-frequency categories (Entertainment, Music, Howto & Style, Comedy) to ensure sufficient per-category training data.

3.6 Experiment 4 — Cross-Category Generalization

To assess whether viral patterns transfer across content genres, the four most frequent categories are selected: Entertainment, Music, Howto & Style, and Comedy. For each ordered pair (train category, test category), a separate SVM is trained on all samples from the training category and evaluated on all samples from the test category using the best hyperparameters from Experiment 2. This produces a 4×4 F1 matrix whose diagonal entries represent in-domain performance and whose off-diagonal entries represent cross-domain generalization.

4 Implementation Details

All experiments are implemented in Python 3 using scikit-learn and the sentence-transformers library. Table 1 summarizes the key hyperparameter settings.

Table 1: Implementation settings used across all experiments.

Setting	Value
Unique videos (after deduplication)	6,351
Train / test split ratio	70 / 30, stratified
Viral label threshold	80th percentile (933,053 views), 20% positive
Random state	42 (all steps)
Structured features	9 dimensions
Sentence embedding model	all-MiniLM-L6-v2 (384 dimensions, frozen)
Full feature dimensionality	393 (9 structured + 384 embedding)
PCA components (Exp. 2)	100 (97.33% variance retained)
LR solver	L-BFGS, max iterations = 1000
LR best C	0.1
LR C grid	$C \in \{0.001, 0.01, 0.1, 1, 10\}$
SVM kernel	RBF
SVM best params	$C = 10, \gamma = 0.1$
SVM C grid	$\{0.1, 1, 10\}$
SVM γ grid	$\{\text{scale}, 0.01, 0.1\}$
Cross-validation folds	5-fold stratified
CV scoring metric	F1
Class imbalance handling	Inverse-frequency sample weighting
Leakage prevention	Scaler / PCA fit on training set only

5 Results

5.1 Experiment 1 — Logistic Regression Baseline

Logistic Regression trained on the nine structured features achieves an AUC of 0.599 and an F1 of 0.352. Cross-validation selected $C = 0.1$ as the best regularization parameter. Table 2 shows the learned coefficient weights sorted by absolute magnitude.

Table 2: Logistic Regression feature coefficients (sorted by absolute value). Positive values indicate association with virality.

Feature	Coefficient	Direction
category id	−0.308	category-dependent
tag count	+0.236	more tags → more viral
title word count	−0.178	shorter titles more viral
publish weekday	+0.164	later in week more viral
description length	+0.119	longer descriptions more viral
title has exclamation	+0.112	“!” associated with virality
title has question	−0.066	“?” negatively associated
publish hour	−0.059	
title length	+0.001	negligible

Content category is the single strongest predictor, consistent with the large variation in viral

rates across categories observed in Figure 1. Tag count ranks second: videos with more tags tend to go viral more often, likely because thorough tagging reflects creator experience and improves discoverability within the recommendation system. Description length carries a similar interpretation — more effort invested in the description signals a more professional production.

A notable finding concerns punctuation in titles. The presence of an exclamation mark is positively associated with virality (+0.112), while a question mark is negatively associated (−0.066). This contrast reflects a difference in content style: exclamation marks appear in emotionally-driven, entertainment-oriented titles (“You Won’t Believe This!”), while question marks appear in informational or tutorial titles (“How do you fix this?”), which tend to belong to lower-viral-rate categories such as Howto & Style and Education.

The model achieves a recall of 0.570, identifying 57% of viral videos, but at low precision (0.255), indicating a substantial false-positive rate. The AUC of 0.599 is modestly above random chance, establishing the ceiling of structured features alone and motivating the semantic features in Experiment 2. Figure 2 shows the ROC curve and confusion matrix.

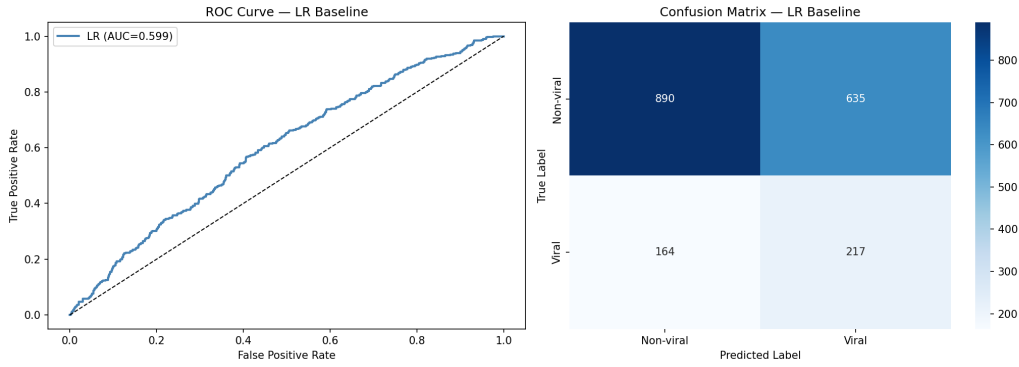


Figure 2: Logistic Regression baseline: ROC curve (left) and confusion matrix (right) on the held-out test set.

5.2 Experiment 2 — Kernel SVM

Adding 384-dimensional sentence embeddings of the video title and tags to the nine structured features, reducing to 100 PCA components, and training an RBF Kernel SVM yields substantial improvement across all metrics. Cross-validation selected $C = 10$ and $\gamma = 0.1$.

Table 3: Test-set performance of the two models.

Model	Accuracy	Precision	Recall	F1	AUC
LR Baseline (structured only)	0.581	0.255	0.570	0.352	0.599
Kernel SVM (structured + embeddings)	0.732	0.375	0.514	0.434	0.711
Absolute improvement (Δ)	+0.151	+0.120	−0.056	+0.082	+0.112
Relative improvement (%)	+26.0%	+47.1%	−9.8%	+23.3%	+18.7%

AUC rises from 0.599 to 0.711 (+18.7%) and F1 from 0.352 to 0.434 (+23.3%). Accuracy and precision improve substantially. Recall decreases slightly ($0.570 \rightarrow 0.514$): the LR baseline, faced with a class-imbalanced problem, casts a wide net and flags many videos as viral, achieving high recall at the cost of low precision (0.255). The SVM, operating in a richer feature space, learns a more selective decision boundary — it misses slightly more true viral videos but is far less likely to misclassify non-viral ones, yielding much higher precision (0.375) and a better overall F1. This trade-off is appropriate for a task where false positives (predicting viral incorrectly) carry their own cost. The AUC of 0.711 means that when given one viral and one non-viral video drawn at random, the model ranks the viral one higher 71.1% of the time — a meaningful result given that the prediction relies solely on pre-publication information.

The gap between the two models suggests that semantic content contributes useful predictive signal, although a stricter ablation study would be needed to fully separate the effect of embeddings from the effect of the more flexible nonlinear SVM model. The structured features alone (AUC 0.599) capture operational signals such as category, tag count, and timing, while the full model additionally encodes *what* the title says.

Figure 3 shows the ROC comparison, SVM confusion matrix, and a side-by-side metric bar chart.

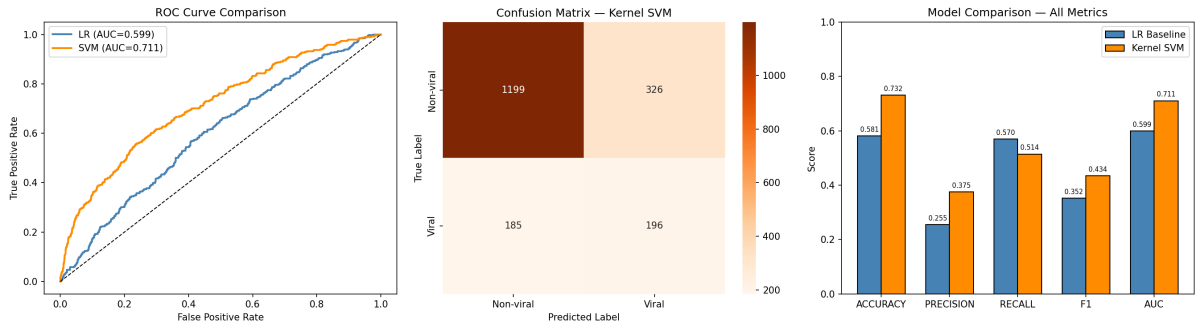


Figure 3: Experiment 2 results: ROC curve comparison (left), SVM confusion matrix (center), and per-metric bar chart comparing LR and SVM (right).

5.3 Experiment 3 — PCA Visualization

Projecting the 393-dimensional training features to 3 principal components retains 46.3% of total variance (PC1: 20.6%, PC2: 15.3%, PC3: 10.4%). The dense, high-rank structure of sentence embeddings means that no small number of principal components can capture the full feature space, so some class overlap in the projected view is expected. Figure 4 shows the 3D projection from two viewing angles: viral and non-viral videos overlap considerably, consistent with the inherent difficulty of the task, but viral videos are not uniformly distributed — they concentrate toward one end of PC1, confirming that the learned features carry genuine discriminative signal even if it is not concentrated in just two or three directions.

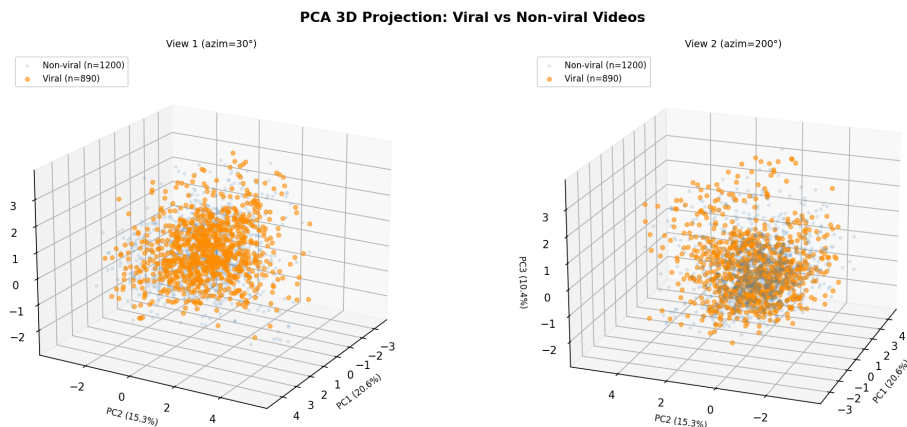


Figure 4: PCA 3D projection of the full training set shown from two viewing angles. Orange points are viral videos; blue points are a subsample of 1,200 non-viral videos (low opacity to serve as background). The two classes overlap considerably but viral videos are not uniformly distributed across the space.

The per-category breakdown in Figure 5 reveals meaningful variation across genres. Gaming and Entertainment exhibit clearer visual separation between viral and non-viral clusters in 3D, while Music videos cluster tightly regardless of virality — consistent with the difficulty of predicting viral Music content from title features alone, and with the weak cross-domain transfer observed for Music in Experiment 4.

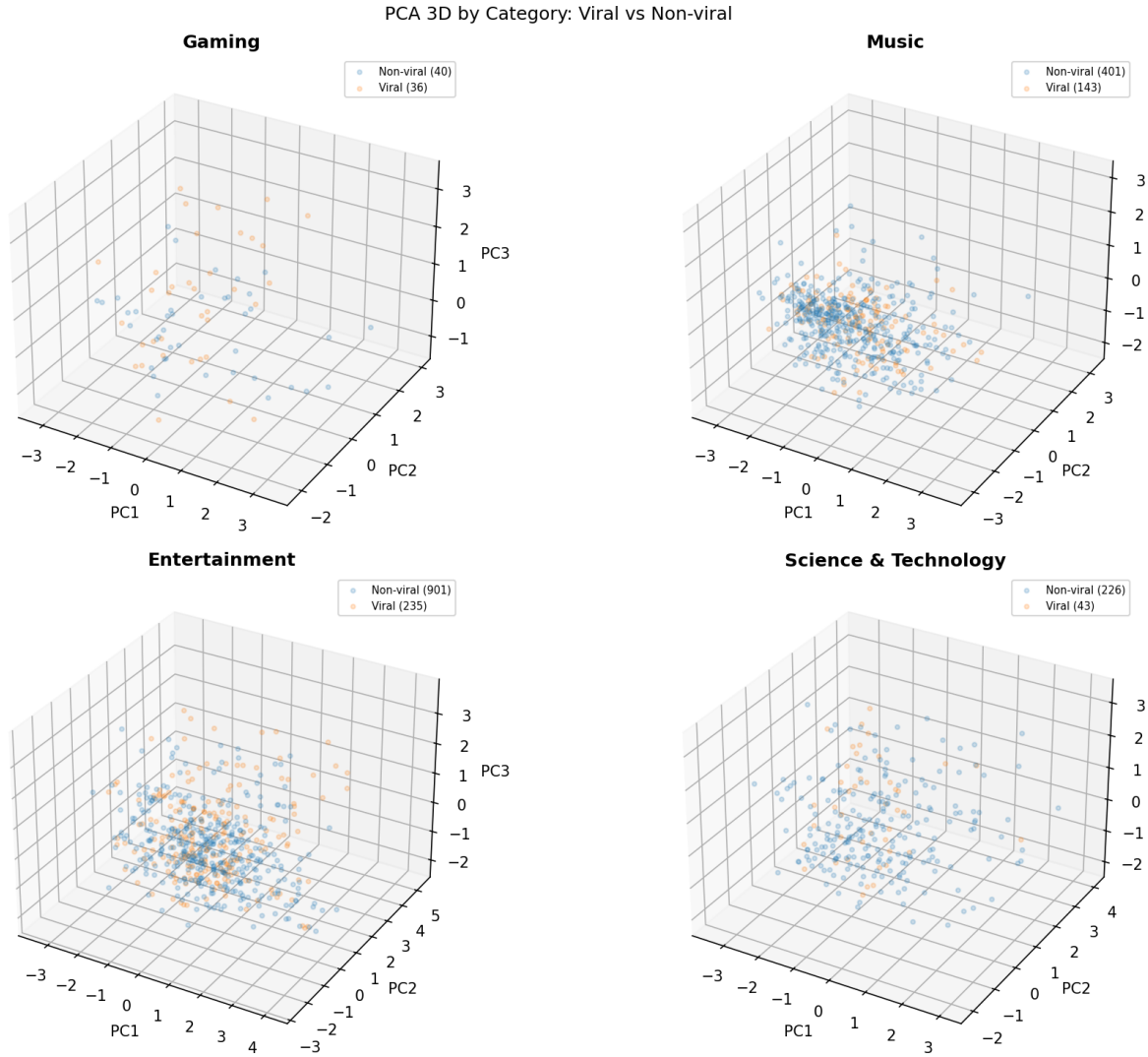


Figure 5: Per-category PCA 3D projections for Gaming, Music, Entertainment, and Science & Technology. Gaming and Entertainment show clearer separation; Music shows the least.

5.4 Experiment 4 — Cross-Category Generalization

Training and testing separate SVM models within and across the four most frequent categories (Entertainment, Music, Howto & Style, Comedy) yields the 4×4 F1 matrix in Figure 6.

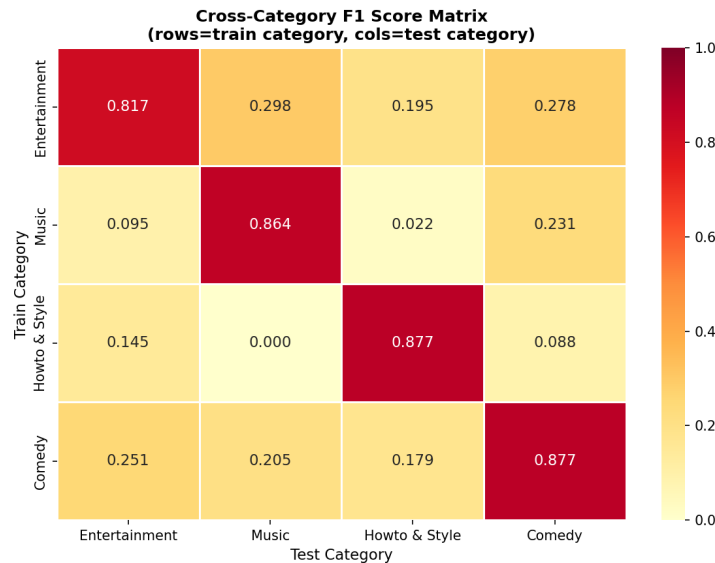


Figure 6: Cross-category F1 matrix. Rows indicate the training category; columns indicate the test category. Diagonal entries are in-domain results; off-diagonal entries are cross-domain results.

The diagonal (in-domain) entries average 0.859, confirming that category-specific models learn strong within-genre patterns. Individual in-domain F1 scores are high: Comedy 0.877, Howto & Style 0.877, Music 0.864, and Entertainment 0.817. The off-diagonal (cross-domain) entries average only 0.166, a drop of 0.693. The most extreme case is Howto & Style trained on Music: $F1 = 0.000$, a complete failure.

This large gap reveals that viral patterns are fundamentally category-specific. A model trained on Comedy learns to recognize comedic framing and humor-driven titles; applied to Music, those signals provide no useful information about whether a track will trend. The finding is not that the models are weak — on their own categories they perform very well — but that the concept of “viral” has a different linguistic fingerprint in each genre.

The higher in-domain F1 scores in this experiment should be interpreted carefully: each category-specific model is trained and evaluated on a narrower within-genre distribution, which may be an easier task than the global held-out test setting. The in-domain F1 of 0.859 thus reflects within-category predictability rather than an improvement over the global model.

6 Discussion

Viral video prediction from pre-publication features is feasible. The Kernel SVM achieves an AUC of 0.711 using only pre-publication information — no view counts, likes, or any post-publication engagement signal. This performance is reasonable for a noisy social-media prediction task, especially given that the model relies solely on pre-publication information. The structured-only baseline (AUC 0.599) already exceeds chance meaningfully, and the addition of semantic embeddings pushes performance substantially higher.

Semantic content contributes meaningful predictive signal. The 18.7% AUC gain from adding sentence embeddings suggests that *what* a creator writes in their title and tags carries predictive signal beyond timing and category, though a full ablation study separating embedding contribution from model flexibility would be needed to confirm this precisely. Title wording is a concrete lever available to creators before publication, and optimizing it may help improve predicted reach.

Operational features reveal creator professionalism. Among the structured features, tag count and description length are the second and fifth strongest predictors respectively. Both serve as proxies for how much effort and care a creator invests in their upload workflow. Experienced creators who carefully tag their videos and write detailed descriptions are more likely to produce content that eventually trends, reflecting not just content quality but also platform literacy.

Punctuation in titles carries interpretable signal. Exclamation marks are positively associated with virality while question marks are negatively associated. This asymmetry reflects a stylistic divide between emotionally-charged entertainment content and informational or tutorial content — the former tends to go viral more often. Such findings illustrate that even simple text features, when interpreted carefully, can reveal genuine patterns in audience behavior.

Viral patterns are category-specific, which is itself a useful finding. The cross-category experiment was designed to ask whether a single global model would be sufficient. The answer is clearly no: in-domain F1 averages 0.859 while cross-domain F1 averages only 0.166. This is not a failure of the modeling approach; it is a discovery about the nature of virality on YouTube. Different genres have different linguistic cultures, and what makes a Comedy title go viral has little in common with what makes a Music video trend. Knowing this guides how future systems should be built — with separate, category-aware models rather than a single one-size-fits-all classifier.

Limitations. The category identifier is treated as a raw integer rather than one-hot encoded, which imposes an arbitrary ordinal structure on a nominal variable; one-hot encoding would be more principled. Additionally, the dataset covers only US trending videos from a limited time window, which may limit generalizability to other regions or periods.

Future directions. Natural extensions include training dedicated per-category models to leverage the strong in-domain performance observed here, incorporating channel-level features such as historical viral rate as a proxy for creator reputation, and exploring thumbnail image features using vision models, since the thumbnail is a key driver of click-through rate that this project did not address.

7 Conclusion

This project demonstrates that predicting YouTube video virality from pre-publication information alone is feasible. By combining nine structured metadata features with 384-dimensional sentence embeddings of the video title and tags, and training an RBF Kernel SVM, we achieve

an AUC of 0.711 and an F1 score of 0.434 on a held-out test set — a 23.3% improvement in F1 and an 18.7% improvement in AUC over the structured-only Logistic Regression baseline.

Feature importance analysis on the baseline model reveals that content category is the strongest single predictor, followed by tag count and the presence of exclamation marks in the title. The contrast between exclamation marks (positive) and question marks (negative) highlights a meaningful stylistic divide between viral entertainment content and non-viral informational content.

The cross-category generalization experiment adds a deeper insight: viral patterns are not universal but strongly genre-dependent, with in-domain F1 averaging 0.859 and cross-domain F1 averaging just 0.166. Rather than undermining the project’s success, this finding clarifies the right modeling strategy going forward — category-aware models that respect the distinct linguistic norms of each genre. Together, these results demonstrate that machine learning can extract meaningful structure from simple pre-publication metadata, with practical implications for content strategy, platform recommendation design, and creator tooling.